EXECUTIVE PITCH DECK

Cold-Chain Integrity Advisor

Predictive Route Thermal Risk Modeling & GxP Compliance for Biopharma Logistics

Founders & Development Team	Antigravity Engineering | Serum & Biopharma Advisory Taskforce
Core Innovation	Physics-based Arrhenius MKT degradation modeling + Real-time IoT Highway Telematics
Regulatory Conformance	CDSCO Draft Guidelines 2023, WHO GDP TRS 961 Annex 9, US FDA 21 CFR Part 11
Production Architecture	Frontend: Next.js | Backend: Node.js + Express.js | Database: MongoDB | Telemetry: MQTT/REST
Live Deployment	Vercel Cloud Serverless + GitHub Enterprise Continuous Delivery

PROBLEM STATEMENT

The $35 Billion Vulnerability in Global Pharma Logistics

Temperature-sensitive biologics suffer devastating spoilage due to reactive monitoring

Annual Economic Waste	Over $35 Billion lost globally every year due to in-transit cold-chain excursions and spoilage.
Post-Mortem Blindness	Legacy USB/NFC data loggers only reveal thermal failures after the vaccines have already arrived ruined at hospitals.
Extreme Ambient Climates	Indian highway corridors regularly exceed 42°C-46°C ambient heat, overwhelming standard passive reefers.
Catastrophic Regulatory Liability	A single unmonitored excursion batch forces mandatory CDSCO quarantine, audit failures, and complete batch destruction.
Core Flaw of Status Quo	Logistics providers act reactively. The market requires predictive pre-departure dispatch advisory.

MARKET OPPORTUNITY

Total Addressable Market (TAM, SAM & SOM)

High-growth tailwinds driven by biopharma expansion and mandatory CDSCO 2023 regulations

TAM: Global Cold-Chain Logistics	$382 Billion by 2030 (CAGR: 16.2%), driven by mRNA vaccines, GLP-1 biologics, and cellular therapies.
SAM: India Pharma Cold-Chain Market	$4.8 Billion by 2028 (CAGR: 19.4%), powered by the 'Pharmacy of the World' vaccine manufacturing hub.
SOM: High-Assurance IoT Integrity Advisory	$620 Million beachhead market targeting Tier-1 CDMOs, vaccine manufacturers, and specialized 3PLs.
Regulatory Catalyst	Mandatory CDSCO Good Distribution Practices (GDP) requires continuous tamper-proof digital thermal logging.
Growth Velocity	Indian refrigerated vehicle fleet growing from 15,000 to 45,000+ units over the next 4 years.

THE SOLUTION

Cold-Chain Integrity Advisor: Proactive & Automated

Transforming blind transportation into an intelligent, self-healing biopharma cold chain

Pre-Departure Advisory	Simulates highway surface temperatures, solar irradiance, and traffic jams before the truck leaves the warehouse.
Arrhenius MKT Analytics	Calculates Mean Kinetic Temperature across all points to measure genuine biological shelf-life degradation.
Dynamic OSRM Highway Routing	Calculates real turn-by-turn road waypoints across Indian National Highways (NH 48, NH 44, Western Corridor).
Instant Excursion Mitigation	Automated real-time rerouting to nearest certified highway dry-ice hubs and pre-cooling depots upon cooling failure.
Zero-Trust Digital Auditing	One-click generation of CDSCO & WHO compliant inspection certificates signed with SHA-256 digital digests.

PLATFORM WORKFLOW

End-to-End Operational Workflow

How the advisory engine protects biopharma from origin synthesis to patient administration

Step 1: Order & Dispatch Formulation	Supply chain planner inputs payload type (Covishield, Covaxin, Insulin), origin hub, and destination clinic.
Step 2: Physics-Based Thermal Assessment	Algorithm evaluates ambient heat, corridor delays, insulation R-values, producing an actionable 0-100 Risk Score.
Step 3: Real-Time IoT Telematics Stream	Reefer GPS and calibrated BLE sensors transmit temperature, ambient heat, door-status, and battery health.
Step 4: Autonomous Breach Detection	Immediate alerts triggered via SMS/Email if temperature drifts beyond narrow target limits (e.g. +2°C to +8°C).
Step 5: Automated GxP Delivery Sign-Off	Consignee receives cryptographic batch release certificate verifying unbroken chain-of-custody.

ARCHITECTURE & TECHNICAL MOAT

Production Tech Stack & Competitive Moat

Engineered on the senior-mandated enterprise stack for hyper-scale and strict compliance

Frontend Layer: Next.js (React 18)	Server-side rendering, low-latency telematics dashboards, interactive Leaflet geo-tracking, Chart.js telemetry curves.
Backend Engine: Node.js + Express.js	High-throughput asynchronous REST microservices, real-time WebSockets, and mathematical Arrhenius calculation modules.
Data Persistence: MongoDB Atlas	Time-series collections for sub-second IoT sensor ingest, immutable GxP audit trail logs, and shipment metadata.
IoT Telematics Gateway	Support for MQTT, CoAP, and Cellular 4G/5G GPS beacon ingestion with encrypted TLS 1.3 payloads.
Our Technical Moat	Proprietary Indian highway corridor thermal microclimate maps + Arrhenius biological degradation modeling.

REGULATORY & COMPLIANCE CONFORMANCE

Built for Global Audits & CDSCO Mandates

Eliminating manual paper logs with tamper-evident, audit-ready compliance certificates

CDSCO Good Distribution Practices 2023	Full adherence to Indian regulatory mandates for continuous cold-chain temperature mapping and documentation.
WHO GDP (TRS 961, Annex 9)	Meets World Health Organization standards for temperature-sensitive pharmaceutical product storage and transit.
US FDA 21 CFR Part 11 Compliance	Electronic records, cryptographic SHA-256 digital verification digests, and immutable chronological audit trails.
ALCOA+ Data Integrity Principles	Attributable, Legible, Contemporaneous, Original, and Accurate telemetry logging.
Instant PDF Audit Export	One-click generation of inspector-ready compliance dossiers with excursion root-cause analysis.

GO-TO-MARKET & TARGET PERSONAS

GTM Strategy & Customer Personas

Targeting high-margin biopharma manufacturers and specialized healthcare 3PLs

Persona 1: Biopharma QA & Regulatory VP	Pain: Audit findings, batch rejections, CDSCO regulatory notices. Value: 100% automated GxP digital sign-off.
Persona 2: Cold-Chain Logistics Director	Pain: In-transit refrigeration breakdowns, driver negligence, dry-ice depletion. Value: Predictive route planning.
Persona 3: Hospital & Depot Chief Pharmacist	Pain: Receiving compromised vaccines without proof of potency. Value: Instant QR-code batch release acceptance.
GTM Phase 1 (Months 1-6): Direct CDMO Outreach	Pilots with Serum Institute, Bharat Biotech, and Zydus Cadila corridor lanes.
GTM Phase 2 (Months 7-18): 3PL Integrations	Strategic partnerships with Blue Dart Medipak, Snowman Logistics, and DHL Supply Chain.

COMPETITIVE BENCHMARKING

Why Cold-Chain Integrity Advisor Wins

Moving past legacy tracking to automated predictive intelligence

Legacy Data Loggers (Sensitech, TempTale)	Passive, post-trip analysis only. Zero pre-trip risk modeling. High manual labor to download data.
Generic Fleet Telematics (Trakmatic, Samsara)	Monitors truck GPS and fuel, but lacks Arrhenius biological MKT modeling and CDSCO GxP compliance templates.
Cold-Chain Integrity Advisor	Holistic solution: Pre-trip route climate simulation + Real-time IoT telematics + Automated GxP audit certificate generation.
Speed-to-Value	Zero-friction customer onboarding portal allowing enterprise workspace setup in under 60 seconds.
Cost Advantage	70% lower total cost of ownership compared to expensive proprietary hardware lock-in.

BUSINESS MODEL & UNIT ECONOMICS

B2B SaaS Subscription + Usage Revenue

Predictable recurring revenue with exceptional gross margins

Tier 1: Growth Tier ($1,499 / mo)	Up to 50 active monitored shipments/month, core predictive route advisory, email excursion alerts, PDF audit certs.
Tier 2: Enterprise Biopharma ($4,999 / mo)	Unlimited shipments, dedicated SLA, custom ERP/SAP integration, custom telematics hardware webhooks.
Per-Consignment Metered Add-On ($25 / shipment)	Single-trip high-value biologic audit certificate generation for ad-hoc 3PL contractors.
Gross Margin Profile	84% gross margins enabled by serverless microservices and automated server-side PDF generation.
Projected ARR	Reaching $2.4M ARR by Month 18 with 45 enterprise biopharma accounts.

EXECUTION ROADMAP

18-Month Product & Growth Roadmap

From live prototype to national standard for biopharma cold chains

Q1-Q2: Prototype & Pilot Validation (Completed)	FastAPI simulator deployed, customer onboarding portal built, Leaflet routing active, ReportLab audit certs verified.
Q3: Production Stack Port (Next.js + Express + Mongo)	Migrating to enterprise stack: Next.js frontend, Node.js/Express backend, MongoDB Atlas time-series database.
Q4: Hardware Agnostic Sensor SDK	Plug-and-play integrations with Teltonika, Queclink, and CalAmp BLE/MQTT temperature gateways.
Q5: AI Autonomous Rerouting Engine	Reinforcement learning for real-time dynamic rerouting around severe Indian highway weather disruptions.
Q6: Pan-India National Rollout	Expansion across 500+ distribution routes covering every state medical supply depot.

The Ask & Future Outlook

Ensuring every single dose of life-saving medicine reaches the patient at 100% potency

Strategic Funding Ask	Seeking $750,000 Seed Investment / Enterprise Pilot Sponsorship to accelerate commercialization.
Use of Funds	45% Core Software Engineering & Stack Migration | 30% Hardware Testing & CDSCO Validation | 25% GTM & CDMO Sales.
Target Milestones	100,000 Monitored Biopharma Shipments, 0 Uncaught Temperature Excursions, 50 Enterprise Clients within 18 Months.
Long-Term Vision	To establish the universal digital trust layer for sovereign healthcare and vaccine resilience across the Global South.
Contact & Demo Access	Live Platform: Vercel Cloud | GitHub: patil2303/ColdChain-Integrity-Advisor | Lead: Shreyas Patil